

Towards an Open Knowledge Graph from Public Data Catalogs

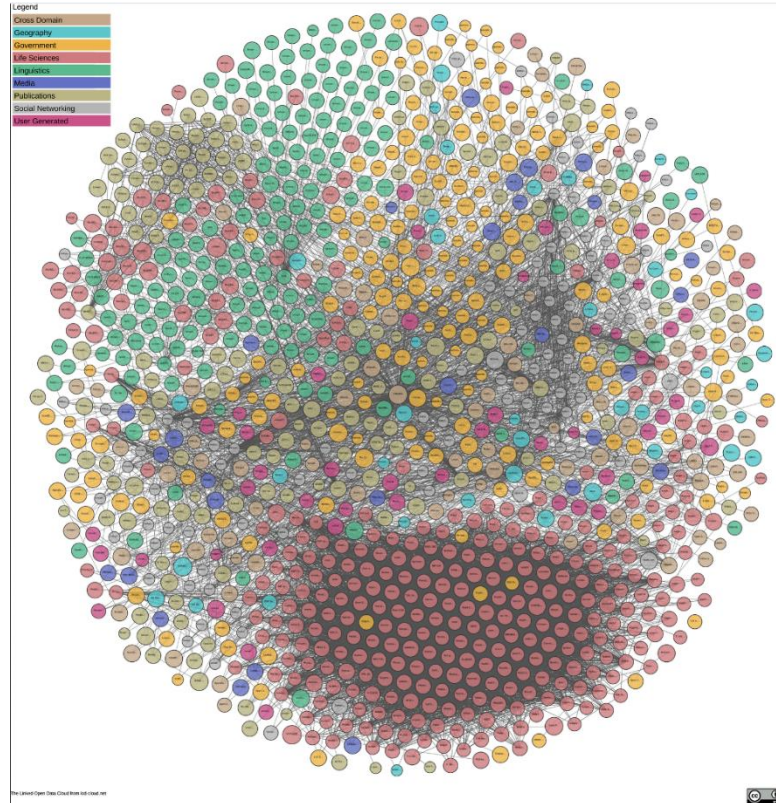
Open Data in *Space* and *Time*



Sebastian Neumaier

Joint work with **Axel Polleres** *and* **Jürgen Umbrich**

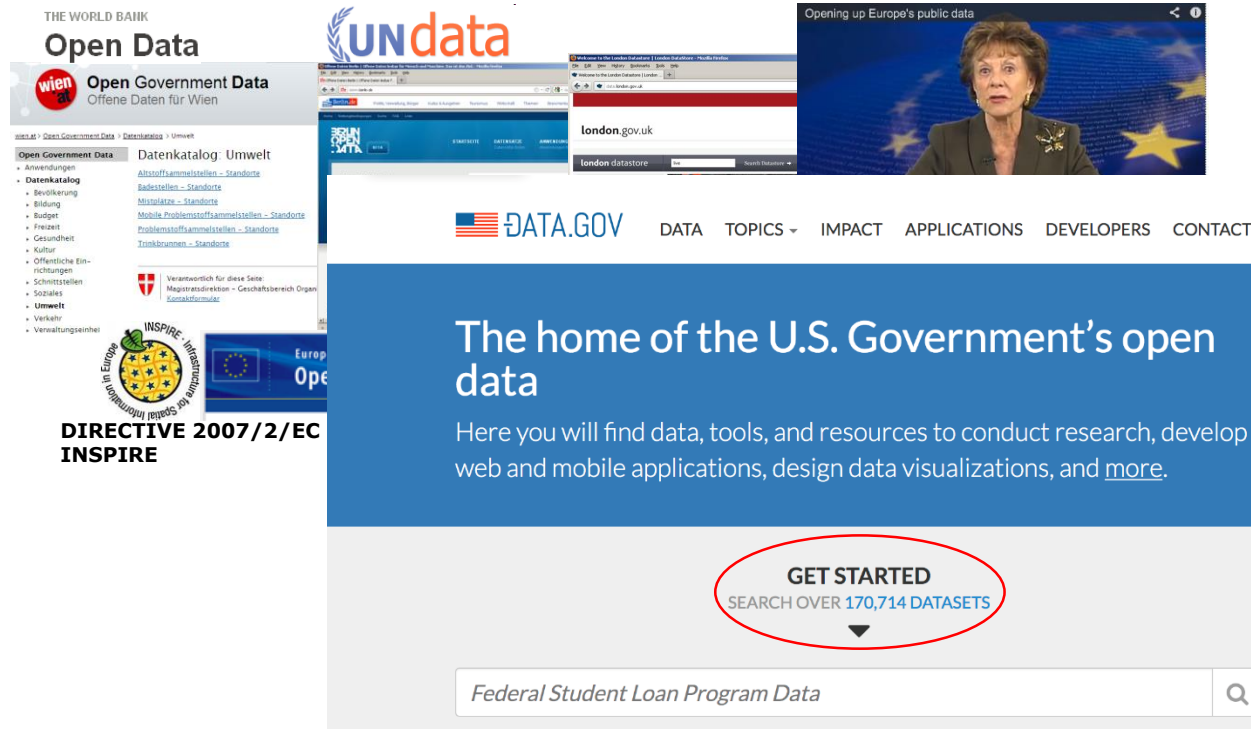
When we hear about Open Data and Knowledge Graphs many think about Linked Open Data...



The Linked Open Data Diagram from lod-cloud.net
Latest release 04-30-2018: **1184 Datasets**

But: Open Data is more than Linked Open Data...

EU & Austria, but also the (previous) US and UK administration are/were pushing Open Data!



The collage features several open data portals and initiatives:

- THE WORLD BANK Open Data**: A portal for open government data.
- wien.at Open Government Data**: A portal for open government data for Vienna.
- UNdata**: A portal for open data from the United Nations.
- london.gov.uk**: A portal for open data from the London Government.
- DATA.GOV**: The U.S. Government's open data portal, featuring a search bar and a "GET STARTED" button.
- INSPIRE**: A portal for open data from the European Union, featuring a search bar and a "GET STARTED" button.

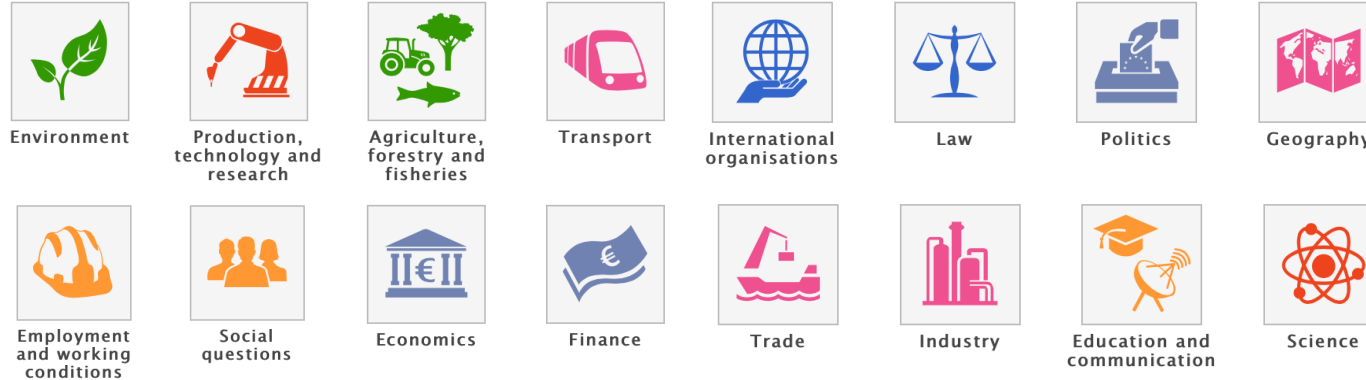
The U.S. Government's open data portal (DATA.GOV) is highlighted with a red circle around the "GET STARTED" button, which says "SEARCH OVER 170,714 DATASETS". Below the button is a search bar containing the text "Federal Student Loan Program Data".

Open Data as a Global Trend

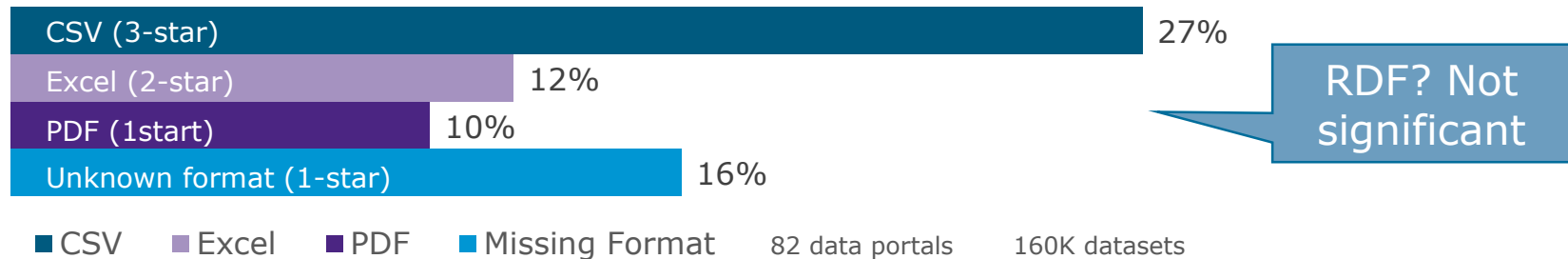
Country	URL	Datasets
United States	data.gov	170.7k
Canada	open.canada.ca	79.1k
UK	data.gov.uk	45.1k
France	www.data.gouv.fr	34.2k
Russia	opengovdata.ru	30.3k
Japan	data.go.jp	21k
Italy	dati.gov.it	20.4k
Germany	govdata.de	19.8k

Data portals of the G8 countries

Data comes in various ways



- Available data is only partially structured and not linked [1]:



What do you find on Open Data Portals?

The screenshot shows the data.gv.at search interface. The search term 'Leopoldstadt' is entered in the search bar. The results section shows 'Suchergebnis zu "Leopoldstadt" (0 gefunden)' and 'Seite 1 von 0'. Below this, there is a section for 'Suchergebnisse von opendataportal.at (0 gefunden)' with a table header. The table has columns: 'Titel', 'Veröffentlichende Stelle / Datenverantwortliche', 'Veröffentlicht auf', 'Letzte Änderung auf', 'Format', and 'Lizenz'. The first row of the table is highlighted in green.

Titel	Veröffentlichende Stelle / Datenverantwortliche	Veröffentlicht auf	Letzte Änderung auf	Format	Lizenz
Stelle	Stelle	opendataportal.at am	opendataportal.at		



Not too much!

Open Data Search is hard...

Why is Search in Open Data a problem?

- No natural language „cues“ like in Web tables...
- Existing knowledge graphs don't cover the domain of „Open Data“ well
- Open Data is not properly geo-referenced

In this talk:

2 approaches how **knowledge graphs** could help

1. Hierarchical labelling of numeric data
2. Hierarchical labelling of spatio-temporal entities

Example Table

<i>federal state</i>	<i>district</i>	<i>year</i>	<i>sex</i>	<i>population</i>
Upper Austria	Linz	2013	male	98157
Upper Austria	Steyr	2013	male	18763
Upper Austria	Wels	2013	male	29730
...	...	...	...	...

Open Data CSVs look more like this

<i>NUTS2</i>	<i>LAU2_NAME</i>	<i>YEAR</i>	<i>SEX</i>	<i>AGE_TOTAL</i>
AT31	Linz	2013	1	98157
AT31	Steyr	2013	1	18763
AT31	Wels	2013	1	29730
...	...		...	...

Why not use the numeric values?

- Identifying the most likely semantic label for a bag of numerical values
- Deliberately ignore surroundings

<i>NUTS2</i>	<i>LAU2_NAME</i>	<i>YEAR</i>	<i>SEX</i>	<i>AGE_TOTAL</i>
AT31	Linz	2013	1	98157
AT31	Steyr	2013	1	18763
AT31	Wels	2013	1	29730
...	...		...	...

Why not use numeric values?

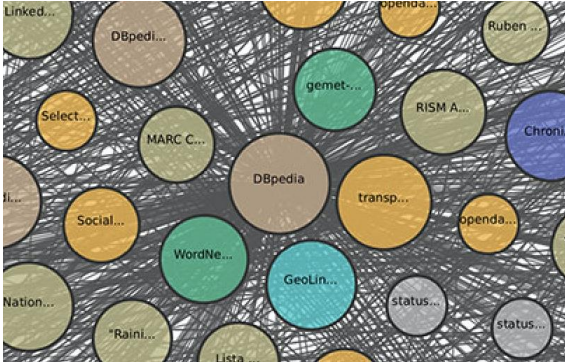
- Identifying the most likely semantic label for a bag of numerical values
- Deliberately ignore surroundings

population (a district) (country Austria)



98157
18763
29730
...

Background Knowledge Graph

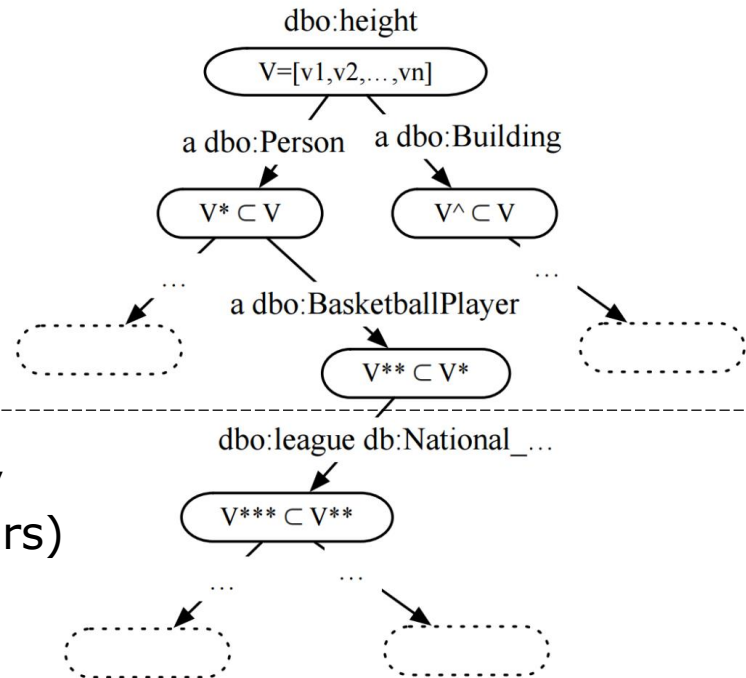


What's in there?

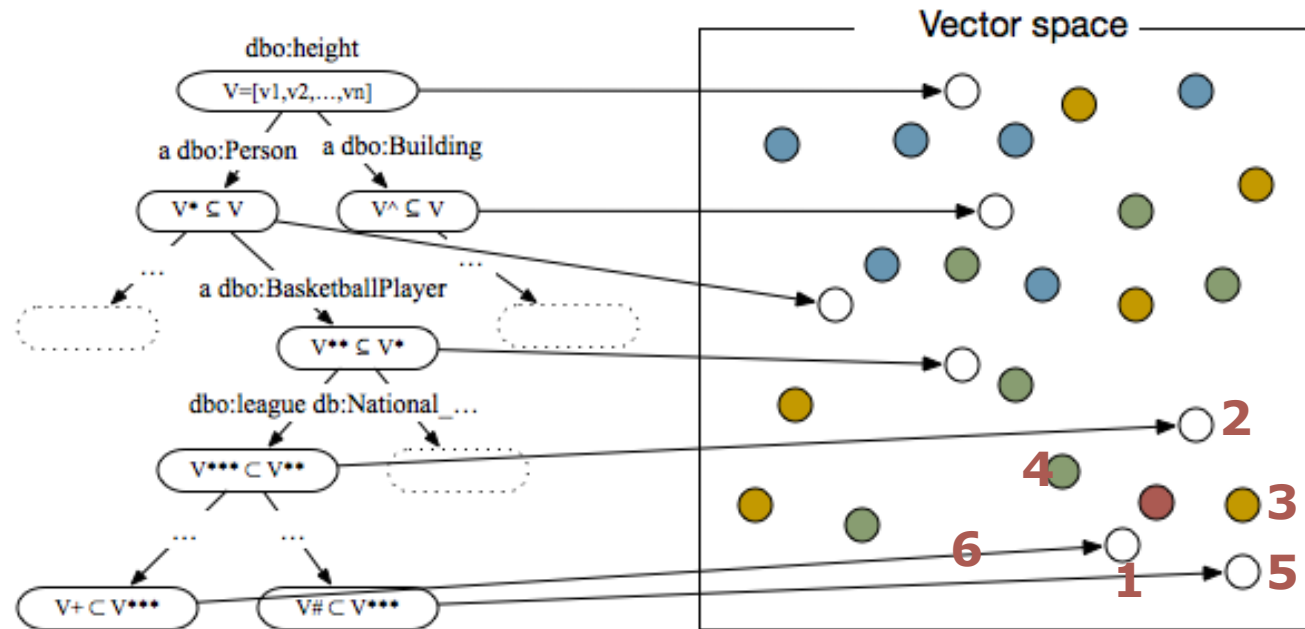
- Cities
 - **Population**
 - **Area**
 - Country
 - Location (**Coordinates**)
 - Economic indicators
 - ...
- Organisations:
 - **Revenues**
 - Board members
 - ...
- Persons (e.g. celebrities, sports)
 - Name
 - Profession
 - **Height**
- Landmarks (e.g. famous buildings)
 - **Country**
 - **Location**
 - **Height**
- Events
 - **Dates**
 - **Location**

Background Knowledge Graph

- Find properties with **numerical range**
- Hierarchical clustering approach
- Two hierarchical layers:
 - Type** hierarchy (using OWL classes)
 - Property-object** hierarchy (shared property-object pairs)



Label based on Nearest Neighbors



Example OD Labelling

populationTotal (a Settlement)
populationDensity (a City)



NUTS1	NUTS2	NUTS3	DISTRICT_CODE	T	WV	WK	BZ	SPR	WBER	ABG.	UNG.	OEVP	SPOE	FPOE	GRUE	BZOE	NEOS
AT1	AT13	AT130		1	9	0	0	0	1163061	503284	9386	81974	136391	89963	103249	1516	44891
AT1	AT13	AT130		2	9	1	0	0	111279	52674	774	9344	12395	6482	14154	114	5412
AT1	AT13	AT130		2	9	2	0	0	98379	51785	646	10324	10236	4700	15398	124	6569
AT1	AT13	AT130		2	9	3	0	0	110527	45483	810	5317	13304	7816	10944	115	3613
AT1	AT13	AT130		2	9	4	0	0	229521	84387	1953	10097	27922	21091	11631	256	5299
AT1	AT13	AT130		2	9	5	0	0	212262	97755	1806	18703	25314	16613	19333	324	9175
AT1	AT13	AT130		2	9	6	0	0	175288	82790	1321	17560	19059	11765	18996	242	8389
AT1	AT13	AT130		2	9	7	0	0	225805	88410	2076	10629	28161	21496	12793	341	6434
AT1	AT13	AT130	90301	3	9	1	3	0	57528	27320	412	4938	6586	3567	6969	68	2789
AT1	AT13	AT130	90401	3	9	1	4	0	21000	11027	138	2401	2253	1068	3082	26	1277
AT1	AT13	AT130	90501	3	9	1	5	0	32751	14327	224	2005	3556	1847	4103	20	1346

- We can assign fine-grained semantic labels
 - **If there is enough evidence in BK**
- *However:* Missing domain knowledge for labelling OD

Future work:

- Complementary to existing approaches (column header labeling, entity linking and relation extraction)
- Combined approaches may improve results
- Focusing on *core dimensions* of *specific domains* e.g. city data, maybe more promising than “general” value labeling.

International Semantic Web conference 2016:

Multi-level semantic labelling of numerical values

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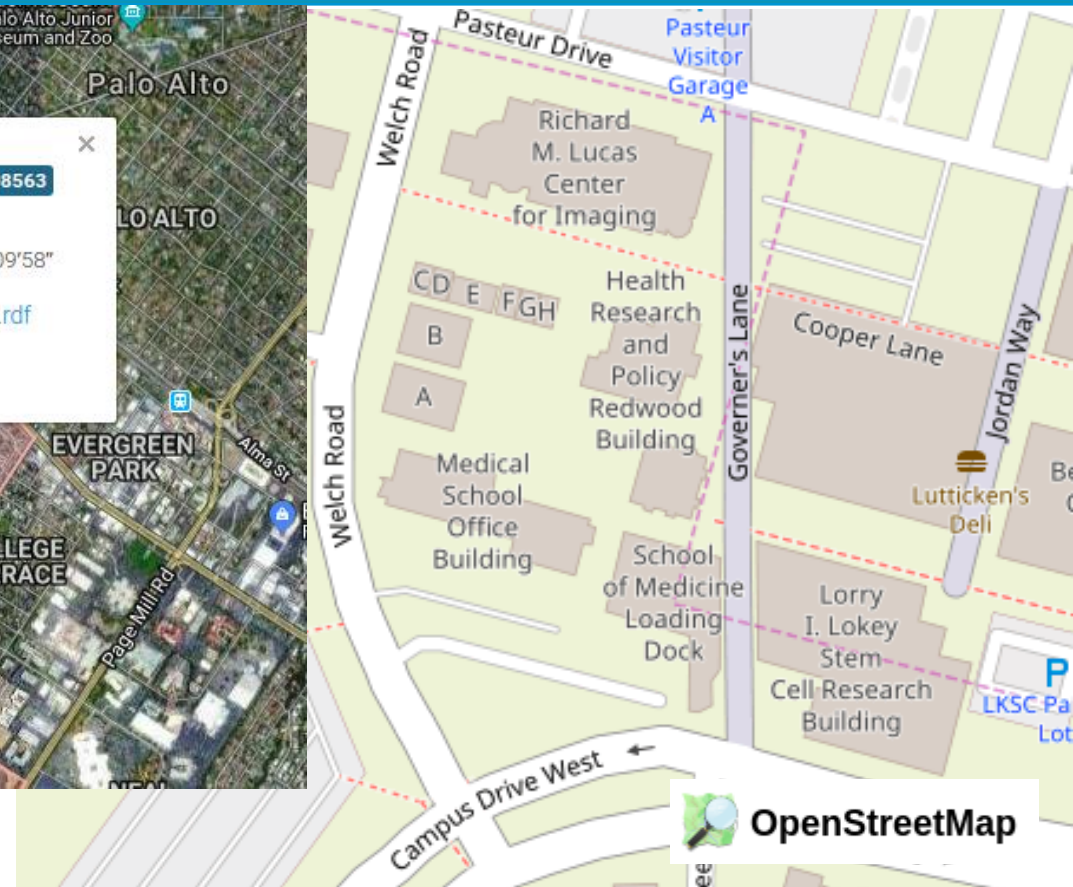
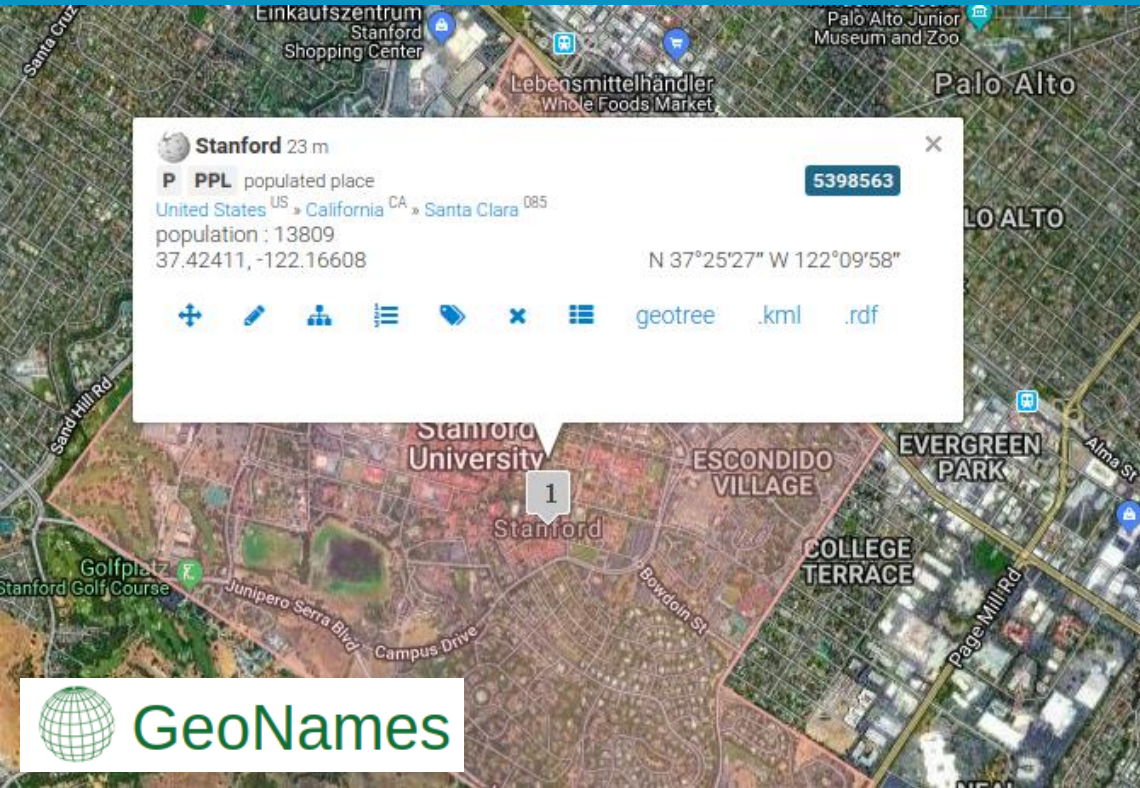
What else can we do/use?

Focus on specific dimensions:

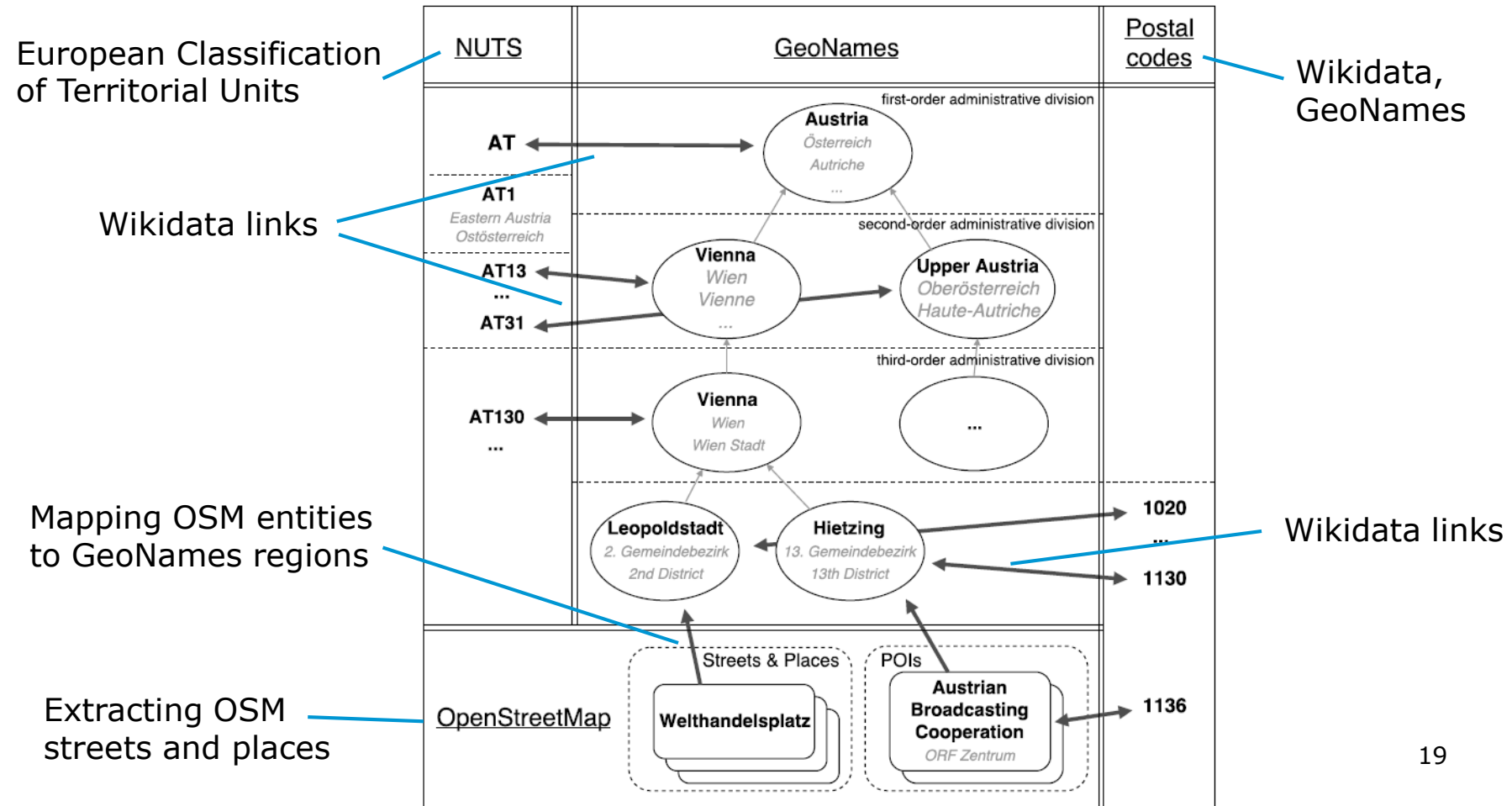
- Particularly **temporal** and **geospatial** queries require better support [2]

<i>NUTS2</i>	<i>LAU2_NAME</i>	<i>YEAR</i>	<i>SEX</i>	<i>AGE_TOTAL</i>
AT31	Linz	2013	1	98157
AT31	Steyr	2013	1	18763
AT31	Wels	2013	1	29730
...	...		...	...

Available Geospatial Knowledge Bases



Geo-Knowledge Graph Construction



Available Temporal Knowledge

 Wikidata Query

Beispiele

Hilfe

Werkzeuge



```
1 SELECT ?itemLabel ?countryLabel ?startLabel ?endLabel
2 WHERE
3 {
4   ?item wdt:P31 wd:Q3558349 ;
5         wdt:P17 ?country ;
6         wdt:P580 ?start ;
7         wdt:P582 ?end .
8   SERVICE wikibase:label { bd:serviceParam wikibase:language "[A
9 }
```


WIKIDATA

3 Ergebnisse in 193 ms

</> Code

Download

itemLabel	countryLabel	startLabel	endLabel
Kabinett Lincoln	Vereinigte Staaten	1861-03-04T00:00:00Z	1865-04-15T00:00:00Z
Presidency of Cristina Fernández de Kirchner	Argentinien	2007-12-10T00:00:00Z	2015-12-09T00:00:00Z
Presidency of Fidel V. Ramos	Philippinen	1992-06-30T00:00:00Z	1998-06-30T00:00:00Z

Periods

Viewing 4226 - 4250 of 5134

Show

25

 periods at a time.

Previous

12...169170171...205206

▲ Label	Earliest start	Latest stop
Tairona Period	900	1600
Taisho Era	1912	1926
Taishō period, 1912-1926	1912	1926
Taizong	976	997
Taizong Liao dynasty	926	947



Temporal Knowledge Graph Construction

```
CONSTRUCT {
  ?event rdfs:label ?label ; dcterms:isPartOf ?Parent ; dcterms:coverage ?geocoordinates ;
  timex:hasStartTime ?StartDateTime ; timex:hasEndTime ?EndDateTime ; dcterms:spatial ?geoentity .
} WHERE {
  # find events with (for the moment) English, German, or non-language-specific labels:
  ?event wdt:P31/wdt:P279* wd:Q1190554 . ?event rdfs:label ?label .
  FILTER( LANG(?label) = "en" || LANG(?label) = "de" || LANG(?label) = "" ).
  # restrict to certain event categories, e.g. (for the moment) elections and sports events:
  { # elections      #sports competitions
    { ?event wdt:P31/wdt:P279* wd:Q40231 } UNION { ?event wdt:P31/wdt:P279* wd:Q13406554 }
  }
  { # with a point in time or start end end date
    { ?event wdt:P585 ?StartDateTime . FILTER ( ?StartDateTime > "1900-01-01T00:00:00"^^xsd:dateTime ) }
    UNION
    { ?event wdt:P580 ?StartDateTime . FILTER ( ?StartDateTime > "1900-01-01T00:00:00"^^xsd:dateTime )
      ?event wdt:P582 ?EndDateT. FILTER ( DATATYPE(?EndDateT) = xsd:dateTime ) }
  }
  OPTIONAL { ?event wdt:P361 ?Parent }
  # specific spatialCoverage if available
  OPTIONAL { ?event wdt:P276?/(wdt:P17|wdt:P131) ?geoentity }
  OPTIONAL { ?event wdt:P276?/wdt:P625 ?geocoordinates }
  BIND ( if(bound(?EndDateT), ?EndDateT, xsd:dateTime(concat(str(xsd:date(?StartDateTime)),"T23:59:59"))) AS ?EndDateTime )
}
```



- Named events and their labels
- Links to parent periods

```
CONSTRUCT {
  ?P rdfs:label ?label ; dcterms:isPartOf ?Parent ; dcterms:spatial ?geo ;
  timex:hasStartTime ?StartDateTime ; timex:hasEndTime ?EndDateTime .
} WHERE {
  {
    { ?P skos:prefLabel ?label } UNION { ?P skos:altLabel ?label } UNION { ?P rdfs:label ?label }
  }
  ?P time:intervalFinishedBy ?End ; time:intervalStartedBy ?Start.
  OPTIONAL { ?P period:spatialCoverage ?geo }
  OPTIONAL { ?P dcterms:spatial ?geo }
  OPTIONAL { ?P dcterms:isPartOf ?Parent. }
  OPTIONAL{ ?End time:hasDateTimeDescription ?EndTime .
    OPTIONAL{ ?EndTime time:year ?EndYear }
    OPTIONAL{ ?EndTime period:latestYear ?EndYear }
  }
  OPTIONAL{ ?Start time:hasDateTimeDescription ?StartTime .
    OPTIONAL{ ?StartTime time:year ?StartYear }
    OPTIONAL{ ?StartTime period:earliestYear ?StartYear }
  }
  OPTIONAL{ ?Start (!period:aux)+ ?StartYear. FILTER (isLiteral(?StartYear)) }
  OPTIONAL{ ?End (!period:aux)+ ?EndYear. FILTER (isLiteral(?StartYear)) }
}

FILTER( ?StartYear >= "1900"^^xsd:gYear || xsd:integer(?StartYear) >= 1900 ||
  ?EndYear >= "1900"^^xsd:gYear || xsd:integer(?EndYear) >= 1900 )

BIND( xsd:dateTime(concat(str(?StartYear),"-01-01T00:00:00")) as ?StartDateTime )
BIND( xsd:dateTime(concat(str(?EndYear),"-12-31T23:59:59")) as ?EndDateTime ) }
```



- Temporal extent: a single beginning and end date
- Links to the spatial coverage

Dataset Labelling

Metadata descriptions

- Geo-entities in titles, descriptions, organizations
- Restricted to origin country
- Temporal tagging using Heildtime framework [3]

CSV cell value disambiguation

- Row context:
 - Filter candidates by potential parents (if available)
- Column context:
 - Least common ancestor of the spatial entities

[3] Strötgen, Gertz: Multilingual and Cross-domain Temporal Tagging. Language Resources and Evaluation, 2013.

Metadata

Tourismus - Ankünfte und Nächtigungen in Oberösterreich

Ankünfte und Nächtigungen in den oberösterreichischen Meldegemeinden ab dem Jahr 2000

Daten und Ressourcen

Ankünfte und Nächtigungen in OÖ seit dem Jahr 2000 Entdecke

Veröffentlichende Stelle: Land Oberösterreich

Datenverantwortliche Stelle: Land Oberösterreich, Abteilung Statistik

Lizenz: Creative Commons Namensnennung 3.0 Österreich

Link zur Lizenz: <https://creativecommons.org/licenses/by/3.0/at/deed.de>

Attributbeschreibung: NUTS2 => Bundesland Oberösterreich; Gemeindenummer bzw. Gemeindename => Erhebungsgemeinde; Jahr => Kalenderjahr; Erhebungsgemeinde lt. Tourismus-Statistik-Verordnung 2002 §2 Abs.7: Städte und Gemeinden mit mehr als 1.000 Gästenächtigungen im Kalenderjahr

CSV

NUTS2	Gemeindename	Jahr	Ankuenfte	Naechtigungen
AT31	Linz	2000	340880	579683
AT31	Steyr	2000	38726	78644
AT31	Wels	2000	84370	150417
AT31	Altheim	2000	4989	10744
AT31	Aspach	2000	2637	21316
AT31	Auerbach	2000	484	3541
AT31	Braunau a. Inn	2000	15748	33911

Diagram illustrating disambiguation of the 'Linz' entity:

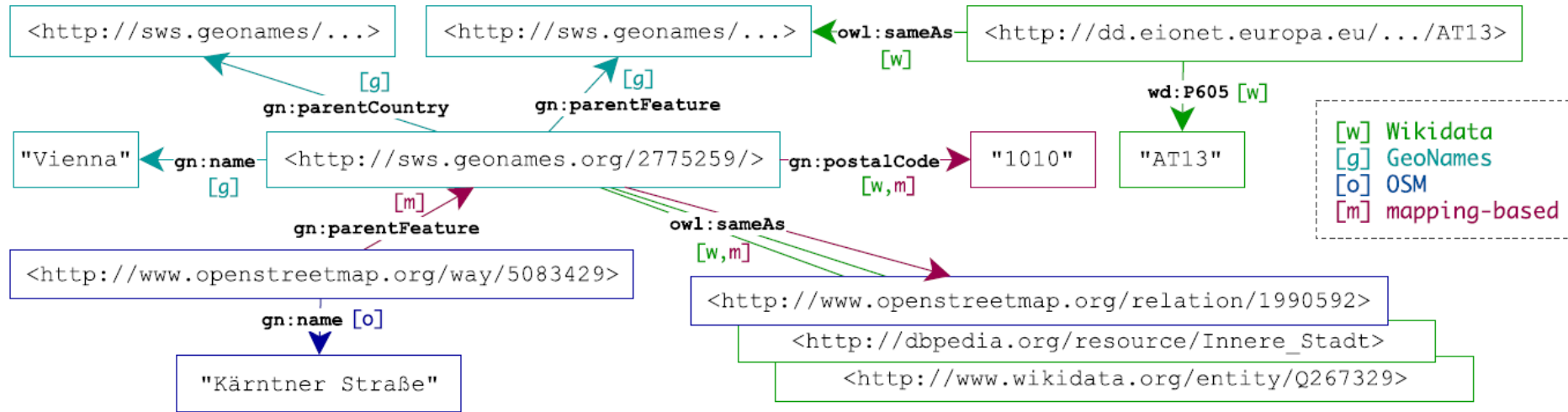
```

    graph LR
        A([Upper Austria  
Oberösterreich  
Haute-Autriche]) --> B([Linz])
        C([Linz]) -- Disambiguate --> D([Linz])
        D --> E([Saxony])
        E --> F([Germany])
    
```

Indexed Datasets

<u>portal</u>	<u>datasets</u>	<u>CSVs</u>	<u>indexed</u>
<i>total</i>			15728
govdata.de	19464	10006	5646
data.gv.at	20799	18283	2791
offenedaten.de	28372	4961	2530
datos.gob.es	17132	8809	1275
data.gov.ie	6215	1194	884
data.overheid.nl	12283	1603	828
data.gov.uk	44513	7814	594
data.gov.gr	6648	414	496
data.gov.sk	1402	877	384
www.data.gouv.fr	28401	6038	258
opengogn.is	54	49	41

RDF Export 1/2: Knowledge Graph



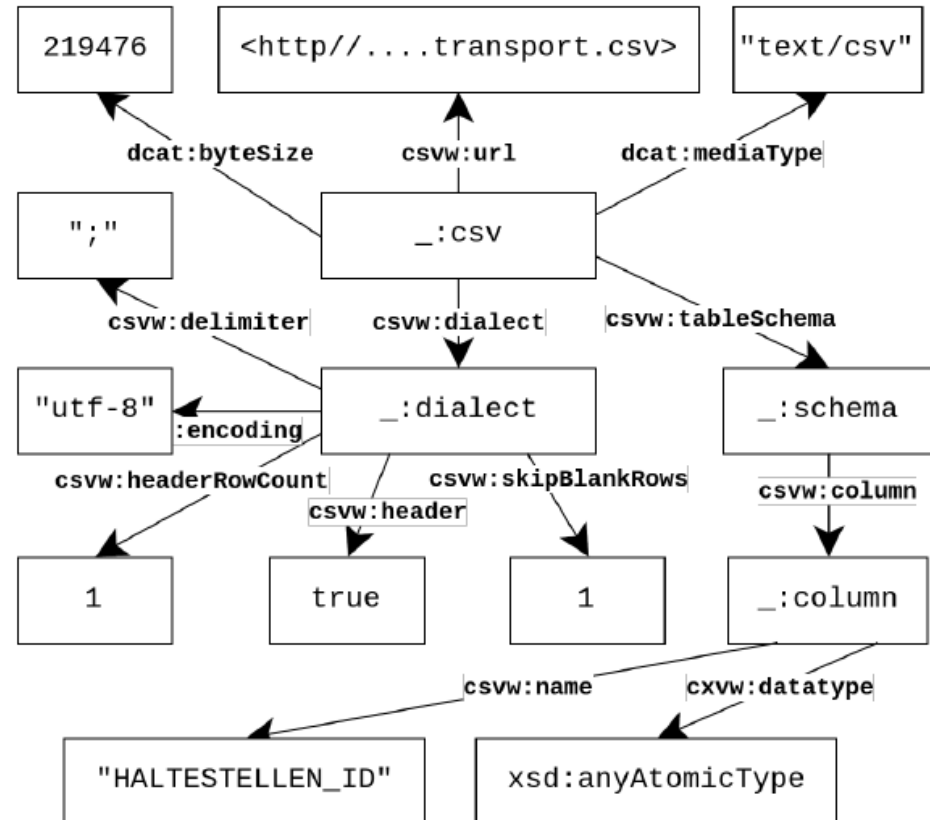
- Spatial and temporal base knowledge graph
- Annotated data points in metadata and CSV cells
- CSV metadata using CSVW vocabulary
 - e.g., delimiter, encoding, header, ...

RDF Export 2/2:

CSV on the Web Metadata [4]

- Note: no real cell level annotations, we needed to add those!
- E.g.:
 - **csvwx:cell**
 - **csvwx:hasTime**
 - **csvw:refersToEntity**
 - ...

Details: cf.:
<http://data.wu.ac.at/ns/csvwx>



SPARQL Endpoint (1)

- Find datasets within time-range and referring to geospatial entity:

```
SELECT ?d ?url WHERE {  
  # select the dates of the past two election in Austria  
  wd:Q1386143 timex:hasStartTime ?t1 .  
  wd:Q19311231 timex:hasStartTime ?t2 .  
  
  # select the min and max date values of a dataset  
  ?d dcat:distribution [  
    dcat:accessURL ?url ;  
    timex:hasStartTime ?start ;  
    timex:hasEndTime ?end  
  ] .  
  # select only datasets about Vienna  
  ?d csvwx:refersToEntity <http://sws.geonames.org/2761369/> .  
  
  FILTER((?start >= ?t1) && (?end <= ?t2))  
}
```

SPARQL Endpoint (2)

- Text search for a time period and its temporal and spatial coverage
- Query for cells within time period and referring to geo-entity

```
SELECT ?d ?url ?rownum WHERE {  
  # get the "Anschluss movement"  
  ?p rdfs:label ?L.  
  FILTER (CONTAINS(?L, "Anschluss movement") ) .  
  ?p timex:hasStartTime ?start ; timex:hasEndTime ?end ; dcterms:spatial ?sp .  
  # find the GeoNames entities  
  ?spatial owl:sameAs ?sp .  
  ?d dcat:distribution [ dcat:accessURL ?url ] .  
  [] csvw:url ?url ; csvw:tableSchema ?s .  
  # find a cell where date falls in the range of the found period  
  ?s csvw:column ?col1 .  
  ?col1 csvwx:cell [  
    csvw:rownum ?rownum ;  
    csvwx:hasTime ?cTime  
  ]  
  FILTER((?cTime >= ?start) && (?cTime <= ?end))  
  # find another cell in the same row where the geo-entity has the  
  # spatial coverage area of the found period as the parent country  
  ?s csvw:column ?col2 .  
  ?col2 csvwx:cell [  
    csvw:rownum ?rownum ;  
    csvwx:refersToEntity [ gn:parentCountry ?spatial ]  
  ]  
}
```

GeoSPARQL Queries

- Standard for representation and querying of geospatial linked data
- (Almost) no complete implementations of GeoSPARQL

```
SELECT ?d ?url ?rownum WHERE {  
  # get the geometry of the Viennese district "Leopoldstadt"  
  <http://sws.geonames.org/2772614/> geosparql:hasGeometry ?polygon .  
  
  ?d dcat:distribution [ dcat:accessURL ?url ] .  
  [ csvw:url ?url ; csvw:tableSchema ?s ].  
  # select the geometries of any annotated cells  
  ?s csvw:column ?col .  
  ?col csvwx:cell [ csvw:rownum ?rownum ; csvwx:refersToEntity [ geosparql:hasGeometry ?g ]  
  
  # filter all annotated data points within the polygon of Leopoldstadt  
  FILTER(geof:sfWithin(?g, ?polygon))  
}
```

Search Interface

Faceted query interface:

- Timespan
- Time pattern
- Geo-entities
- Full-text queries

Back end:

- **MongoDB** for efficient key look-ups
- **ElasticSearch** for indexing and full-text queries
- **Virtuoso** as a triple store

▼ Temporal filters

Filter results by timespan: ☐ Off ☒ Title & description ☐ CSV columns

1/2010 1/2020

Filter pattern

Apply Filter

Linz

Republic of Austria > Oberösterreich > Linz Stadt > Linz

Spatial entity or Full-text results

Hotspot - Standorte - [Hotspot Standorte](#)  Stadt Linz <http://data.gv.at>

POI's (Points of Interest) für Hotspot (freies, kostenloses WiFi) in der Stadt Linz. Die Koordinaten sind im im EPSG-Codes WGS84 verfügbar.

Nummer	Latitude	Longitude	Name	Kurztext	Start im Jahr	Ende im Jahr	Stadt	Postleitzahl
4007	48,304793	14,299414	Hotspot Linz - Rotes...	Hier ist nur einer v...	2013	0	Linz	4020

Finanzgebarung der Gemeinden in Oberösterreich - Oö. Gemeinde- Land Oberösterreich <http://data.gv.at>

Finanzgebarung 2015  Finanzdaten der 444 oberösterreichischen Gemeinden

Jahr	NUTS2	Gemeindenummer	Gemeindename	Ordentliche Einnahme...	Ordentliche Ausgaben	Außero Einnah...
2015	AT31	40101	Linz	628704196,3	718773006,9	131859

- Open (Structured) Data is a rich source of Knowledge worthwhile to tap into
- Most of it is not (yet) Linked Data.

What we did:

- Hierarchical knowledge graph of spatial and temporal entities
- Algorithms to annotate CSV tables and their metadata descriptions

What's next:

- Enable GeoSPARQL (or an alternative geospatial-query language)
- Parsing coordinates in datasets
- Parse other file formats, e.g., XML, PDF, ...
- Test other domains such as tweets or web pages (e.g., newspaper articles)

Thank you!
Questions?



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Backup Slides

Total numbers of spatial and temporal annotations of metadata descriptions and columns:

<u>Columns</u>	<i>Spatial</i> <u>Metadata</u>	<i>Temporal</i> <u>Columns</u>	<u>Metadata</u>
3518	11231	2822	9112

10 random CSV datasets per portal (11 portals), 10 random rows per dataset:

- In total inspected 101 datasets, i.e. 1010 rows
- 87 Correctly assigned labels at the dataset level
- 37 CSV datasets that contain potentially missing annotations (e.g. text that would need to be parsed first, or malformed CSVs, etc.)
- 9 Incorrect links to GeoNames
- 9 Incorrect links to OSM